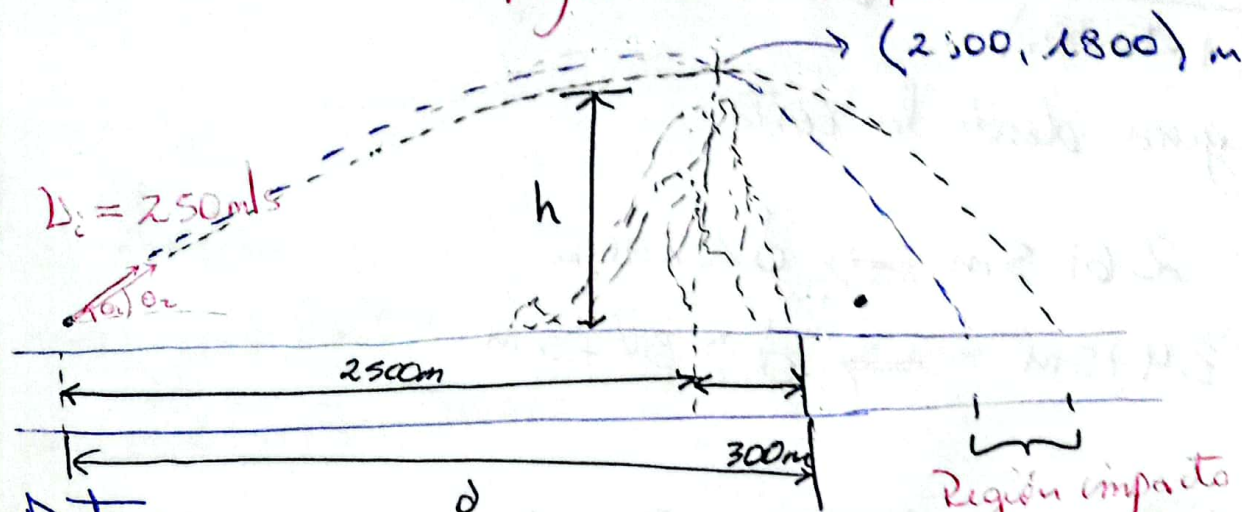


Ayudantía 09/05



Datos

* $V_i = 250 \text{ m/s}$

* $\theta_1, \theta_2 = ?$ Zona impacto

* $(2500, 1800) \text{ m}$ encuentro Trayectoria

Ecuaciones

$x(t) = V_0 \cos \theta t \quad (1)$

$y(t) = V_0 \sin \theta t - \frac{1}{2} g t^2 \quad (2)$

Ecuación de la Trayectoria

$x(t) = V_0 \cos \theta t \Rightarrow t = \frac{x}{V_0 \cos \theta} \quad (3) ; \quad (3) \rightarrow (2)$

$\Rightarrow y_f = V_0 \sin \theta \left(\frac{x_f}{V_0 \cos \theta} \right) - \frac{1}{2} g \left(\frac{x_f}{V_0 \cos \theta} \right)^2$
 $= \tan \theta x_f - \frac{g x_f^2}{2 V_0^2} (1 + \tan^2 \theta) \quad (4)$

$\Rightarrow \frac{g x_f^2}{2 V_0^2} \tan^2 \theta - x_f \tan \theta + y_f = 0 \quad / \cdot \frac{2 V_0^2}{g x_f^2}$

$\tan^2 \theta - \frac{2 V_0^2}{g x_f} \tan \theta + \frac{2 V_0^2 y_f}{g x_f^2} = 0$

Ecuación cuadrática $= \tan \theta_1 = 3.9051 \tan \theta_2 = 1.197$

$\theta_1 = 75.6^\circ \wedge \theta_2 = 50.1^\circ$

Regiones de Impacto para $\tan \theta_1$ y $\tan \theta_2$ con (4), $y_f = 0$

$0 = \tan \theta_1 x_{f-1} - \frac{g x_{f-1}^2}{2 V_0^2} (1 + \tan^2 \theta_1) \quad / \cdot \frac{1}{x_{f-1}}$

$0 = \tan \theta_1 - \frac{g x_{f-1}}{2 V_0^2} (1 + \tan^2 \theta_1) \Rightarrow x_{f-1} = \frac{2 V_0^2 \tan \theta_1}{g (1 + \tan^2 \theta_1)} = 3062 \text{ m}$

$$x_{f-2} = \frac{2H_0^2 \tan \theta_2}{g(1 + \tan^2 \theta_2)} = 6275. \text{ m}$$

Distancia segura desde la costa

$$x_{f-1} - d = 261.5 \text{ m} \Rightarrow d < 270 \text{ m}$$

$$x_{f-2} - d = 3475 \text{ m} \Rightarrow d > 3475 \text{ m}$$

2-



Datos: $R = 1.0 \text{ m}$

$$\omega(0) = 0 \text{ m}$$

1 vuelta (2π) en 2.0 s

Ecuaciones

$$\Delta\theta = \omega_0 t + \frac{1}{2} \alpha t^2 \Rightarrow \alpha = \frac{2\Delta\theta}{t^2} = \pi \text{ rad/s}^2$$

$$(a) \text{ t en 2 vueltas} \Rightarrow \Delta\theta = \frac{1}{2} \alpha t^2 \Rightarrow t = \sqrt{\frac{2\Delta\theta}{\alpha}} = \sqrt{8} \text{ s}$$

$$(b) v = R\omega \hat{\phi} ; \omega = \cancel{\omega_0} + \alpha t = \alpha t$$

$$\Rightarrow v = 6.3 \text{ m/s} \hat{\phi}$$

$$\vec{a} = \vec{a}_t + \vec{a}_c = R\alpha \hat{\phi} + R\omega^2 (-\hat{r})$$

$$= (3.14 \hat{\phi} - 39.5 \hat{r}) \text{ m/s}^2$$

$$3. \quad \hat{r} = \cos\phi \hat{i} + \sin\phi \hat{j}$$

$$\hat{\phi} = -\sin\phi \hat{i} + \cos\phi \hat{j}$$

$$(a) \quad \hat{r} \cdot \hat{r} = 1$$

$$\Rightarrow (\cos\phi \hat{i} + \sin\phi \hat{j}) \cdot (\cos\phi \hat{i} + \sin\phi \hat{j})$$

$$= \cos^2\phi (\hat{i} \cdot \hat{i}) + \cos\phi \sin\phi (\hat{i} \cdot \hat{j}) + \sin\phi \cos\phi (\hat{j} \cdot \hat{i}) + \sin^2\phi (\hat{j} \cdot \hat{j})$$

$$= \cos^2\phi + \sin^2\phi = 1$$

$$(b) \quad \hat{r} \cdot \hat{\phi} = 0$$

$$\Rightarrow (\cos\phi \hat{i} + \sin\phi \hat{j}) \cdot (-\sin\phi \hat{i} + \cos\phi \hat{j})$$

$$= -\cos\phi \sin\phi (\hat{i} \cdot \hat{i}) + \cos^2\phi (\hat{i} \cdot \hat{j}) - \sin^2\phi (\hat{j} \cdot \hat{i}) + \sin\phi \cos\phi (\hat{j} \cdot \hat{j})$$

$$= 0$$

$$(c) \quad \hat{r} \times \hat{r} = 0$$

$$(\cos\phi \hat{i} + \sin\phi \hat{j}) \times (\cos\phi \hat{i} + \sin\phi \hat{j})$$

$$\Rightarrow \cos^2\phi (\hat{i} \times \hat{i}) + \cos\phi \sin\phi (\hat{i} \times \hat{j}) + \sin\phi \cos\phi (\hat{j} \times \hat{i}) + \sin^2\phi (\hat{j} \times \hat{j})$$

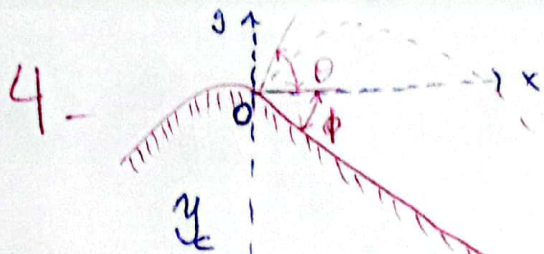
$$= \cos\phi \sin\phi \hat{k} + \sin\phi \cos\phi (-\hat{k}) = 0$$

$$(d) \quad \hat{r} \times \hat{\phi} = \hat{k}$$

$$\Rightarrow (\cos\phi \hat{i} + \sin\phi \hat{j}) \times (-\sin\phi \hat{i} + \cos\phi \hat{j})$$

$$= -\cos\phi \sin\phi (\hat{i} \times \hat{i}) + \cos^2\phi (\hat{i} \times \hat{j}) - \sin^2\phi (\hat{j} \times \hat{i}) + \sin\phi \cos\phi (\hat{j} \times \hat{j})$$

$$\Rightarrow \cos^2\phi \hat{k} + \sin^2\phi \hat{k} = \hat{k}$$



Coordenadas en el punto de lanzamiento $x_0 = y_0 = 0$

Ecuaciones: $x(t) = v_0 \cos \theta t$, $y(t) = v_0 \sin \theta t - \frac{1}{2} g t^2$

$$\Rightarrow t = \frac{x}{v_0 \cos \theta} \rightarrow y_p = x \tan \theta - \frac{g x^2}{2 v_0^2 \cos^2 \theta}$$

pendiente $\Rightarrow \tan \phi = -\frac{y_c}{x} \Rightarrow y_c = -\tan \phi x$

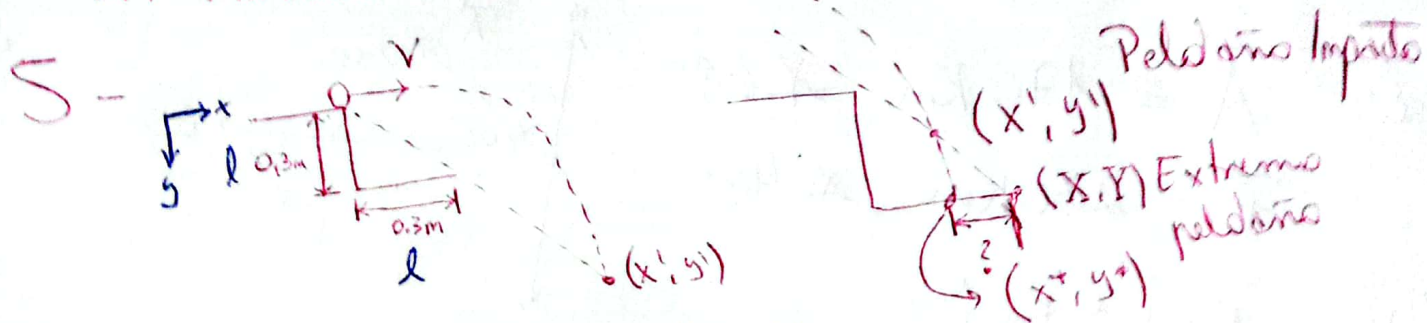
Punto de intersección $y_p = y_c$

$$\Rightarrow -x \tan \phi = x \tan \theta - \frac{g x^2}{2 v_0^2 \cos^2 \theta} \quad / \cdot \frac{1}{x}$$

$$-\tan \phi = \tan \theta - \frac{g x}{2 v_0^2 \cos^2 \theta} \Rightarrow \frac{g x}{2 v_0^2 \cos^2 \theta} = \tan \theta + \tan \phi$$

$$\Rightarrow x = \frac{v_0^2 \cos^2 \theta (\tan \theta + \tan \phi)}{g}$$

1a) Determinar el número de peldaños en el canal.



Datos

$$V_0 = 8.0 \text{ m/s}$$

$$g = 9.8 \text{ m/s}^2$$

$$l = 0.3 \text{ m}$$

Ecuaciones

$$\text{Recta: } y = x \quad \rightarrow V_{0y} = 0$$

$$\text{Parábola: } y = V_{0y}t + \frac{1}{2}gt^2; \quad t = \frac{x}{V}$$

$$y = \frac{gx^2}{2V^2}$$

parabólica - recta

(x', y') punto intersección

$$\begin{aligned} \textcircled{1} \quad y' &= x' \quad / \dots \Rightarrow \textcircled{1} + \textcircled{2} \Rightarrow x' = \frac{g}{2V^2} x'^2 \Rightarrow x' = \frac{2V^2}{g} \\ \textcircled{2} \quad y' &= \frac{g}{2V^2} x'^2 \end{aligned}$$

$$y' = x' = \frac{2V^2}{g} \Rightarrow \frac{y'}{l} = \# \text{ peldaños} \Rightarrow \frac{2V^2}{gl} = \boxed{\frac{43}{N}} \text{ peldaño}$$

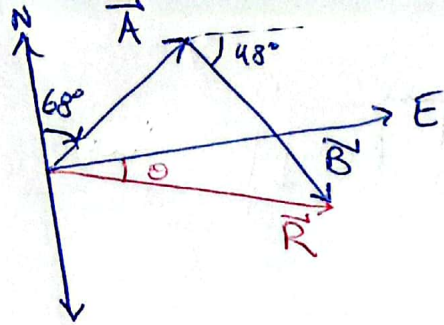
(x^*, y^*) punto impacto en peldaño

$$y^* = \frac{g}{2V^2} x^{*2} \Rightarrow x^* = \sqrt{\frac{2V^2 y^*}{g}} = \sqrt{\frac{2V^2 l \cdot N}{g}} = 12.98 \text{ m}$$

$$\text{Extremo peldaño } X = N \cdot l = 12.9 \text{ m}$$

$$d = X - x^* = \boxed{0.02 \text{ m}}$$

6- $\vec{A} = 170 \text{ Km } \angle 68^\circ$
 $\vec{B} = 230 \text{ Km } \angle 48^\circ$



$$\vec{R} = R_x \hat{i} + R_y \hat{j}$$

$$= (A_x + B_x) \hat{i} + (A_y - B_y) \hat{j}$$

$$R_x = A_x + B_x = 170 \sin 68^\circ + 230 \cos 48^\circ = 311.5 \text{ Km}$$

$$R_y = A_y - B_y = 170 \cos 68^\circ - 230 \sin 48^\circ = -107.2 \text{ Km}$$

$$|\vec{R}| = R = \sqrt{R_x^2 + R_y^2} = 329.4 \text{ Km}$$

$$\theta = \text{Arctan}\left(\frac{R_y}{R_x}\right) = -19^\circ$$

7- $\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$; $\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$
 $\vec{B} = B_x \hat{i} + B_y \hat{j} + B_z \hat{k}$

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} = \hat{i}(A_y B_z - A_z B_y) + \hat{j}(A_x B_z - A_z B_x) + \hat{k}(A_x B_y - A_y B_x)$$

$$\vec{B} \times \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ B_x & B_y & B_z \\ A_x & A_y & A_z \end{vmatrix} = \hat{i}(B_y A_z - B_z A_y) + \hat{j}(B_x A_z - B_z A_x) + \hat{k}(B_x A_y - B_y A_x)$$

$\vec{A} \times \vec{B}$ y $\vec{B} \times \vec{A}$ tienen misma magnitud pero diferente dirección